

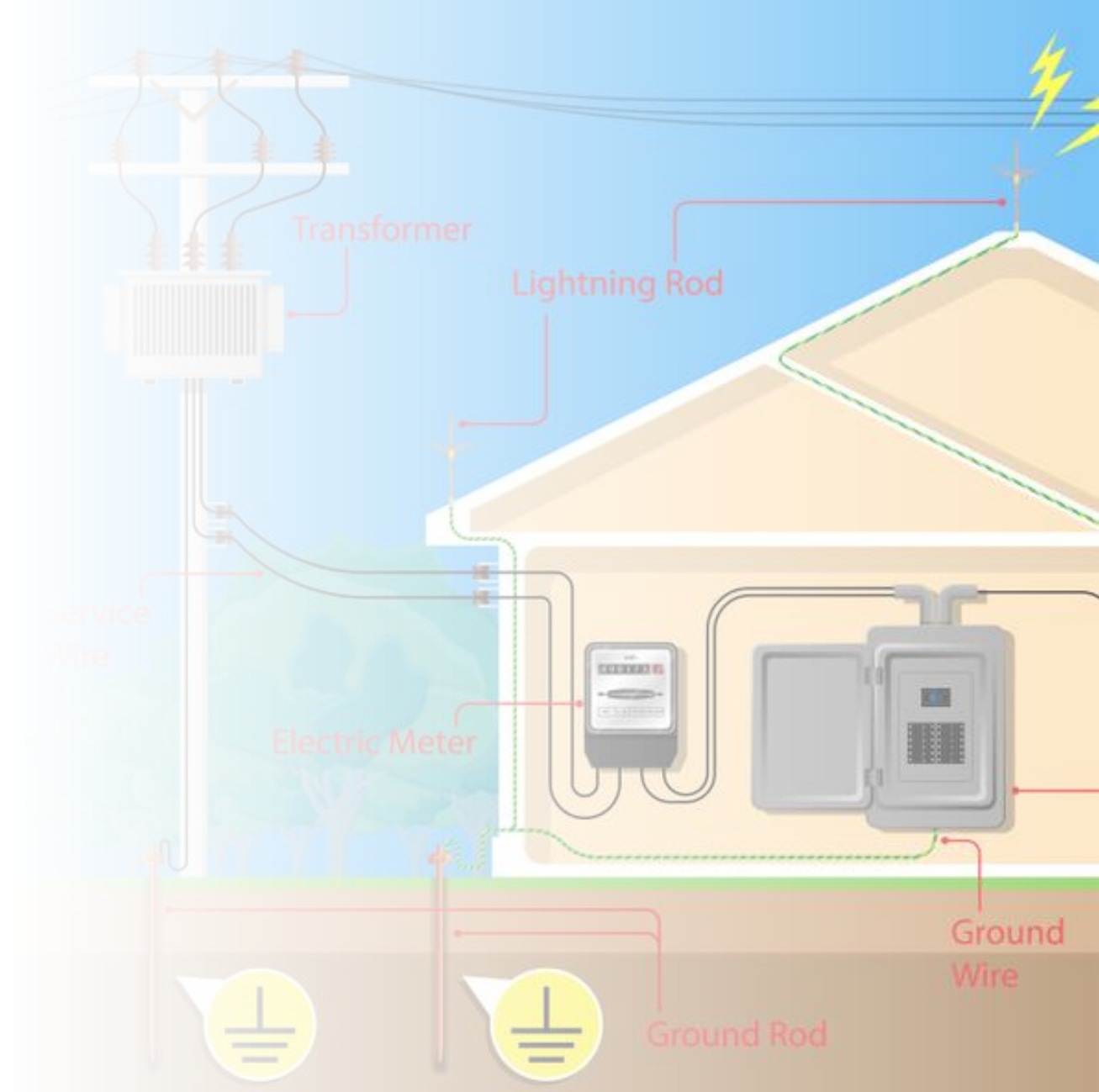
Residential Electricity Demand Forecasting: A 6-Hour Ahead Prediction Model

Group 2

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Introduction & Problem Statement

Presenter: Yang Yang

Short-Term Residential Electricity Forecasting

⚡ Dynamic Nature

Residential electricity usage is highly volatile and historically difficult to predict accurately.

⚡ Our Challenge

Specifically, accurately predicting electricity consumption for the **next 6 hours**.

⚡ Why It Matters

Precise forecasting is critical for efficient **load balancing**, enabling demand response, and driving significant cost reduction.

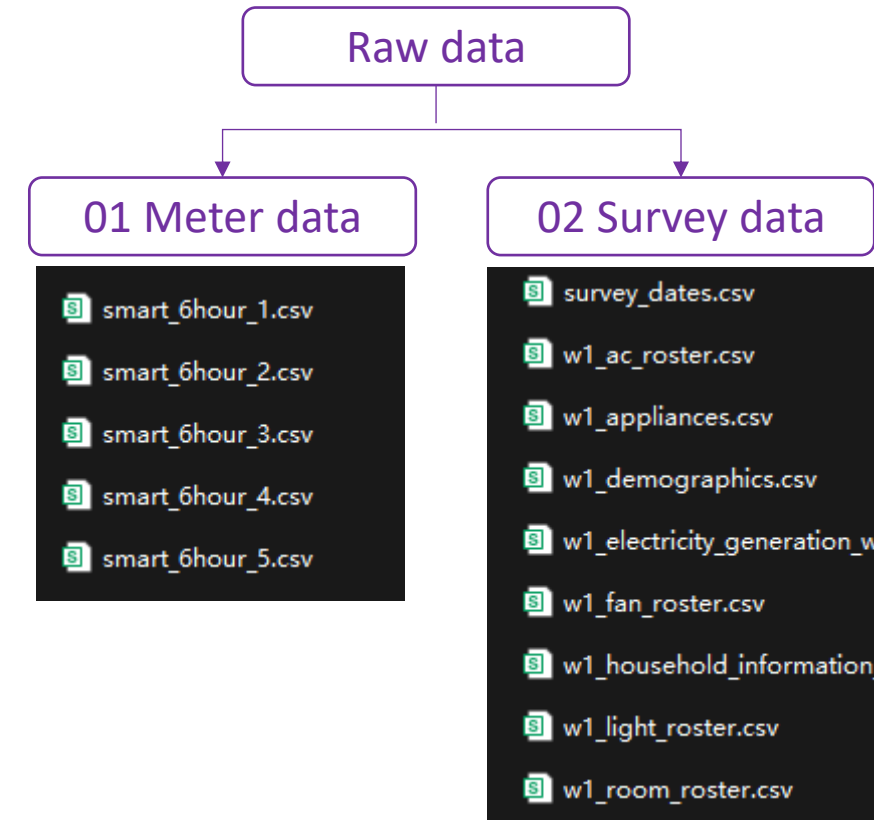
⚡ Project Goal

To define the scope and implement a model that addresses these short-term fluctuations

Dataset Overview

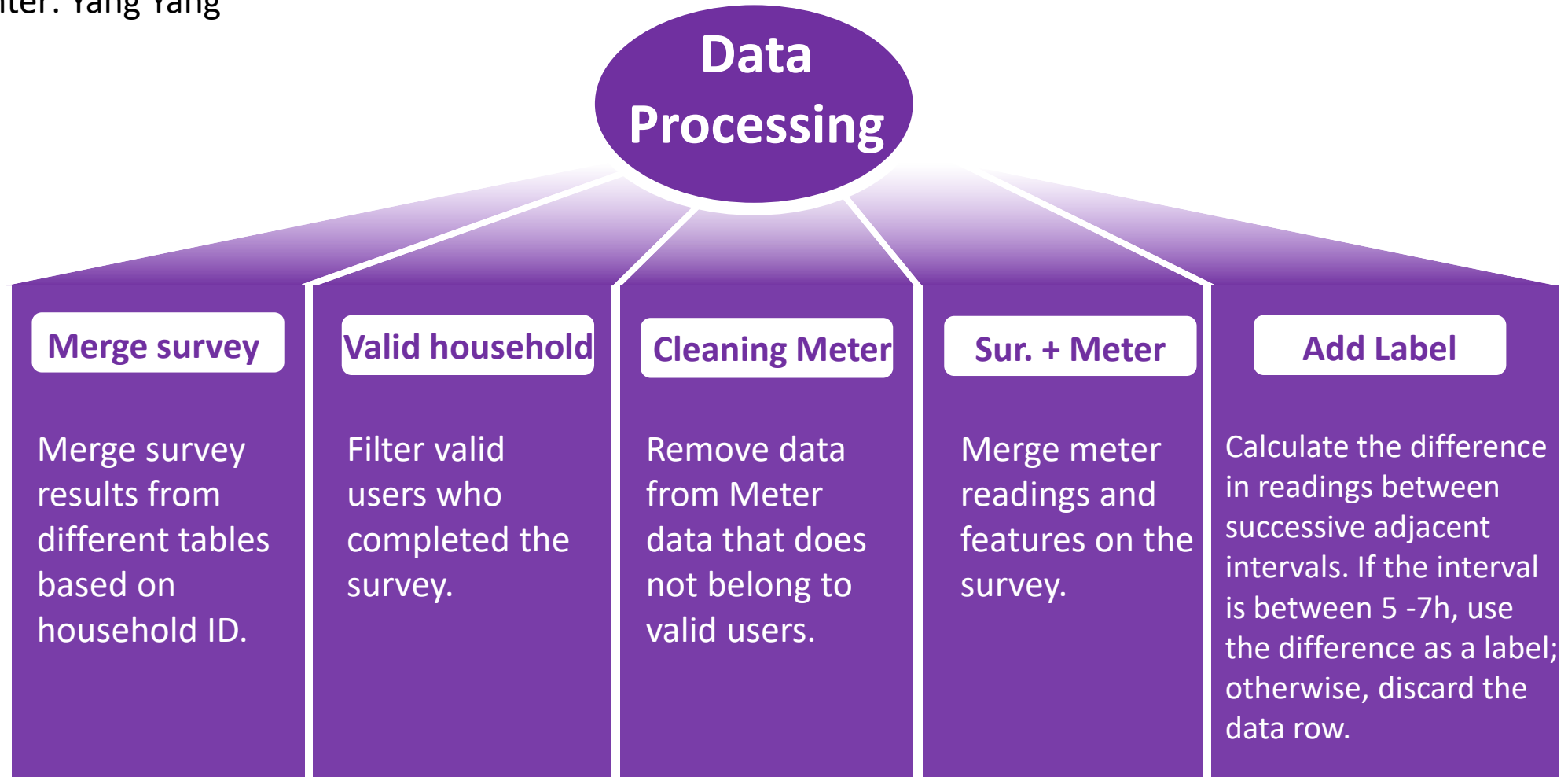
Presenter: Yang Yang

- Dataset from IEEE DataPort
 - <https://ieee-dataport.org/open-access/residential-electricity-consumption-dataset-sri-lanka-recon-sl>
- **Includes:**
 - Smart meter readings (20+ months and 1,438 households from Sri Lanka)
 - Household surveys (demographics, house area, appliance usage, behavior,)
- **Advantages:** Rich & Comprehensive & Real-world significance
- **Challenges:** irregular time intervals, missing values
 - Readings are mostly recorded approximately every 6 hours.
 - Each device records time differently each day



Data Processing & Feature Engineering

Presenter: Yang Yang



Data Processing & Feature Engineering

Presenter: Yang Yang

Each line represents a meter log

- The dataset contains **48 columns**, target is **future_6h_consumption**, features covering:
 - **household_ID**, meter reading **date**, **time**, and **electricity meter readings**
 - Household survey information, including:
 - **Household Demographic Characteristics**
 - **Appliance Ownership and Usage (AC, Fans and Lighting)**
 - **Housing Structure and Physical Attributes**
 - **Household Activities and Socioeconomic Status**
 - **Cooking Energy Sources**
- Valid meter data log: **463399 rows**
- Valid households: **909 households**

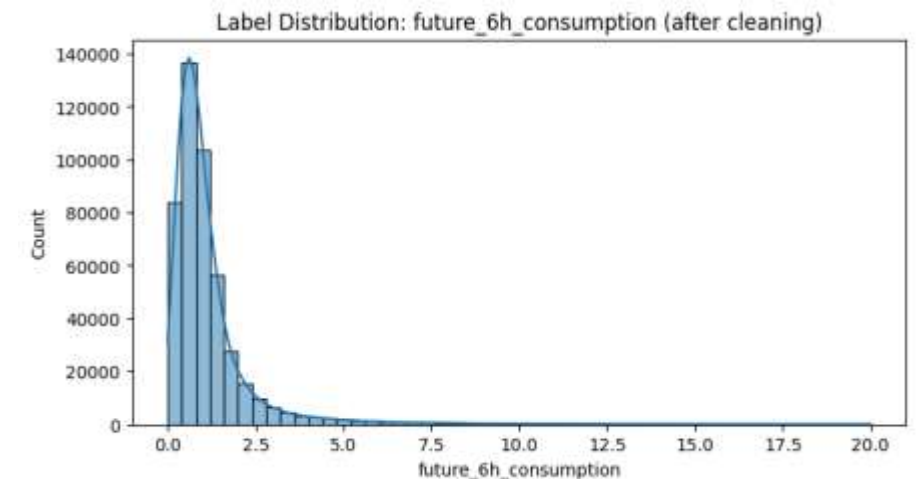


Fig. 1 Label Distribution: future_6h_consumption

Data Processing & Feature Engineering

Presenter: Yang Yang

- Add refactoring features

- Temporal features
 - hour (0-24)
 - dayofweek (1-7)
 - month (1-12)
- Historical Electricity Consumption
 - avg_5d_same_time_kwh
 - avg_1d_same_time_kwh
 - avg_1w_same_time_kwh
 - consumption_per_member
 - backward_avg_consumption

```
future_6h_consumption    1.000000
hour                     0.168210
backward_avg_consumption 0.129860
consumption_per_member   0.111033
avg_1d_same_time_kwh     0.108292
avg_5d_same_time_kwh     0.086168
avg_1w_same_time_kwh     0.059981
month                    0.006408
dayofweek                -0.008201
fan_density              -0.061345
Name: future_6h_consumption, dtype: float64
```


Data Processing & Feature Engineering

Presenter: Yang Yang

- *Split training and test sets.*

- To ensure a fair evaluation of the model's performance, the dataset is split **at the household level**.
- Households in the **training set** do **not** appear in the **test set**.
- A **20% / 80% split** is applied:
 - **20% households** → Test set
 - **80% households** → Training set

This prevents data leakage and ensures the model is tested on **previously unseen households**, improving the reliability of the results.

```
Total households: 909
Training households: 727
Testing households: 182
X_train shape: (367767, 54)
X_test shape: (94946, 54)
Number of overlapping households (should be 0): 0
Saved train and test sets.
```


Linear Regression - Workflow Overview

Presenter: Xi Li

1. Feature Selection & Cleaning

- **Removed irrelevant or ID-like fields** (e.g., household_ID, DATE/TIME, some survey fields).
- Focused on features that are meaningful for **short-term load forecasting**.

2. Feature Engineering

- Created **cyclical time features**: hour_sin/cos, month_sin/cos.
 - **Filled missing values** (numeric → median; categorical → most frequent).
 - Applied **one-hot encoding** to categorical features.
 - Standardized numerical features.
- *Purpose*: Capture daily/monthly cycles and ensure all features operate on comparable scales.

3. Train & Evaluate Linear Models

- Ridge Regression (L2)
- Lasso Regression (L1)
- Elastic Net (L1 + L2)

→ Compare different regularization strategies.

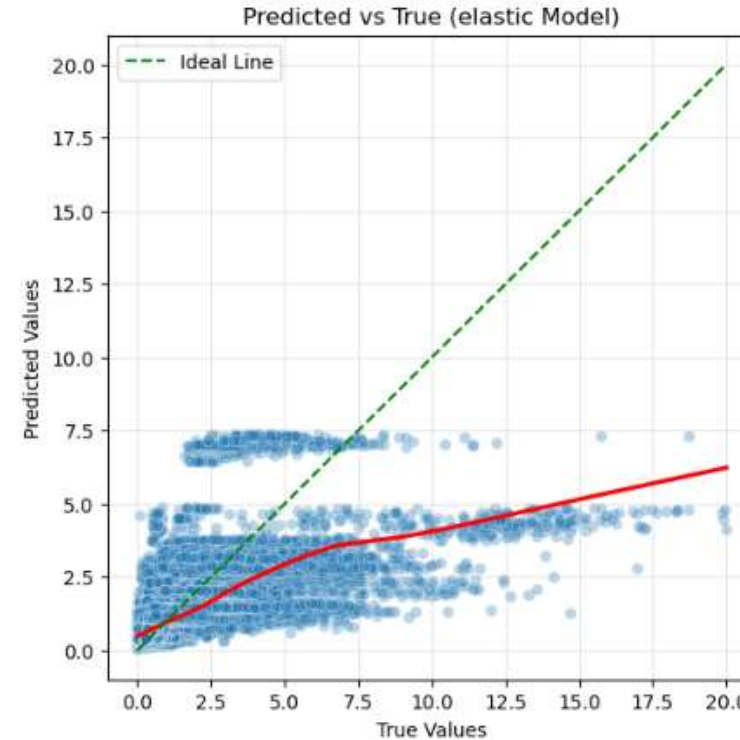
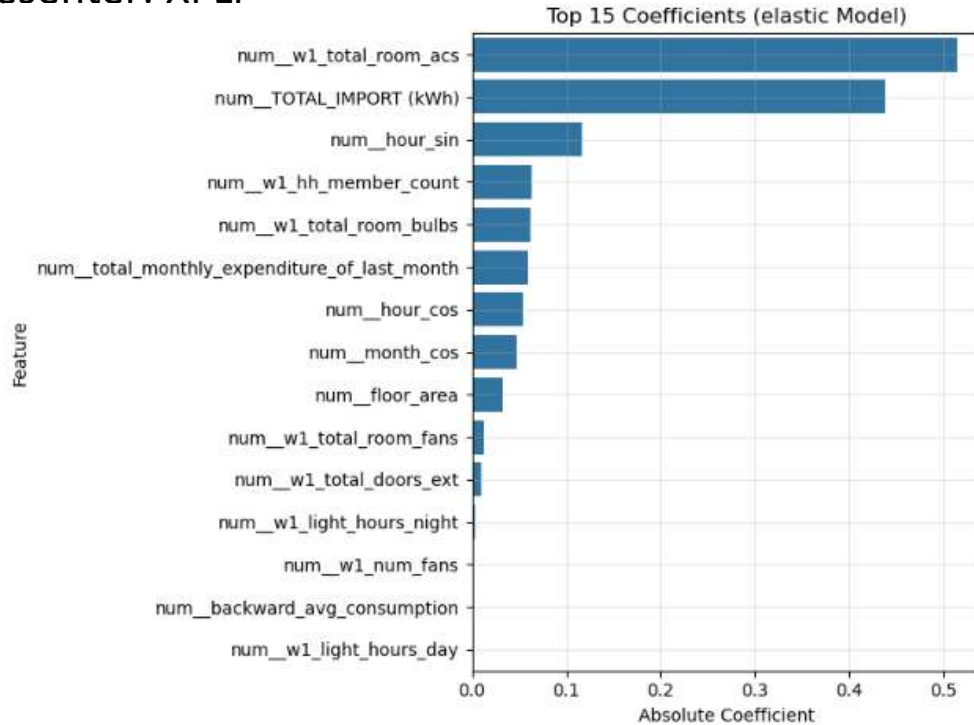
Evaluate on Test Set

- RMSE ≈ 0.86
- MAE ≈ 0.46
- $R^2 \approx 0.42$

→ **Elastic Net performed best** and was selected as the final linear model.

Linear Regression – Key Findings

Presenter: Xi Li



- Linear regression provides a **simple and interpretable baseline**.
- It identifies **AC usage, total import, and temporal features** as key drivers of energy consumption.
- It performs well for **normal usage patterns**, but struggles with **extreme values and non-linear behaviors**.
- These insights guide the design of more advanced models (Random Forest, XGBoost, neural networks).

Random Forest

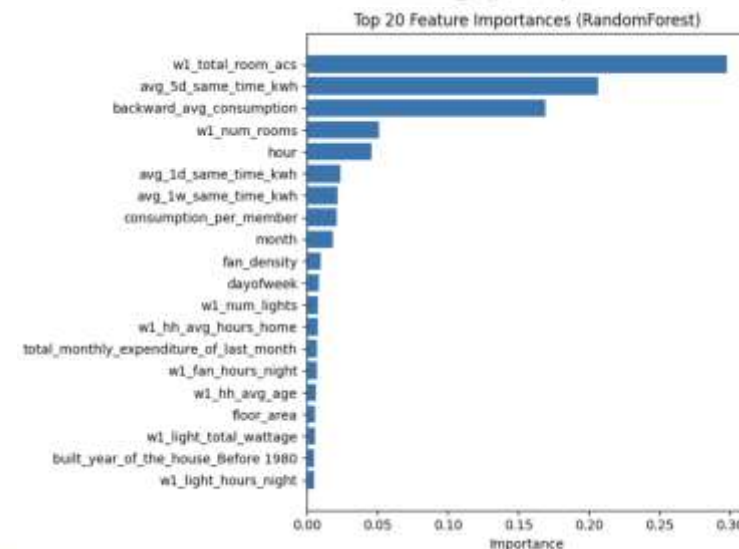
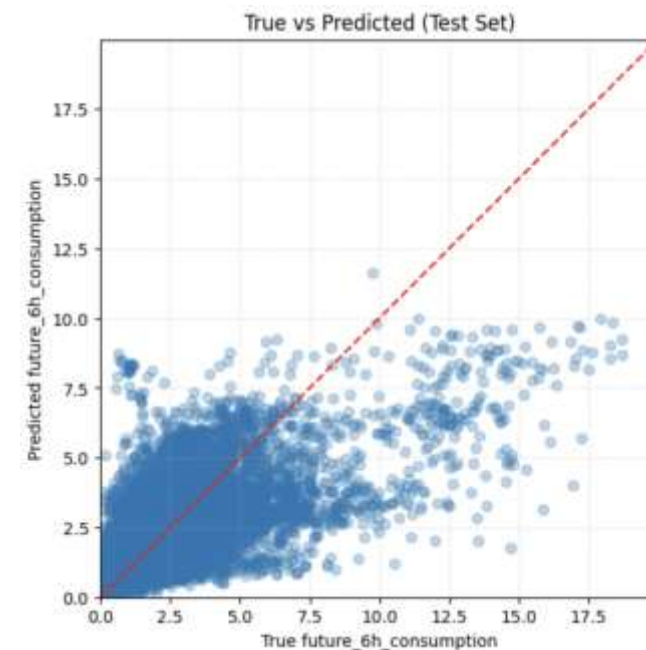
Presenter: Eason Ma

- **Features Used**

- Dropped: household_ID, DATE, TIME, timestamp, TOTAL_IMPORT (kWh)
- **48 features total**
 - Demographics: household size, avg age, #children, #seniors, hours at home, work-out rate
 - House & appliances: rooms, windows/doors, lights, fans, ACs, floor area, monthly expenditure
 - Time: hour of day, day of week, month
 - Historical patterns: backward average consumption, per-member usage, avg_5d_same_time_kwh, avg_1d_same_time_kwh, avg_1w_same_time_kwh, fan density
- Preprocessing: numeric = passthrough, categorical = One-Hot Encoding

- **Random Forest Baseline**

- Model: RandomForestRegressor + preprocessing pipeline
- Final RF config: 200 trees, max_depth = 50, min_samples_split = 5, min_samples_leaf = 2
- Training time: ~47 minutes on Colab CPU
- **Test performance:**
 - MAE $\approx$ **0.3563**
 - RMSE $\approx$ **0.7145**
 - $R^2 \approx$ **0.5991**
- Increasing tree depth/number gave **only small gains** but **large training cost**



XGBoost & Model Comparison

Presenter: Eason Ma

- **XGBoost Setup**

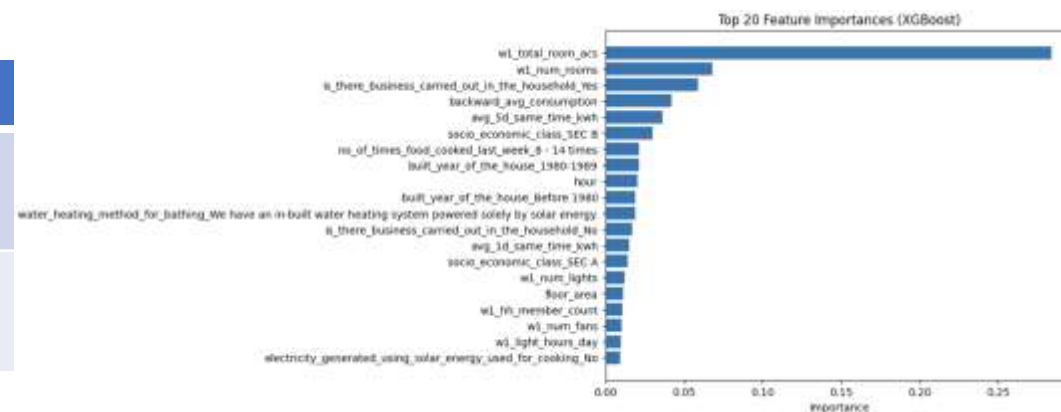
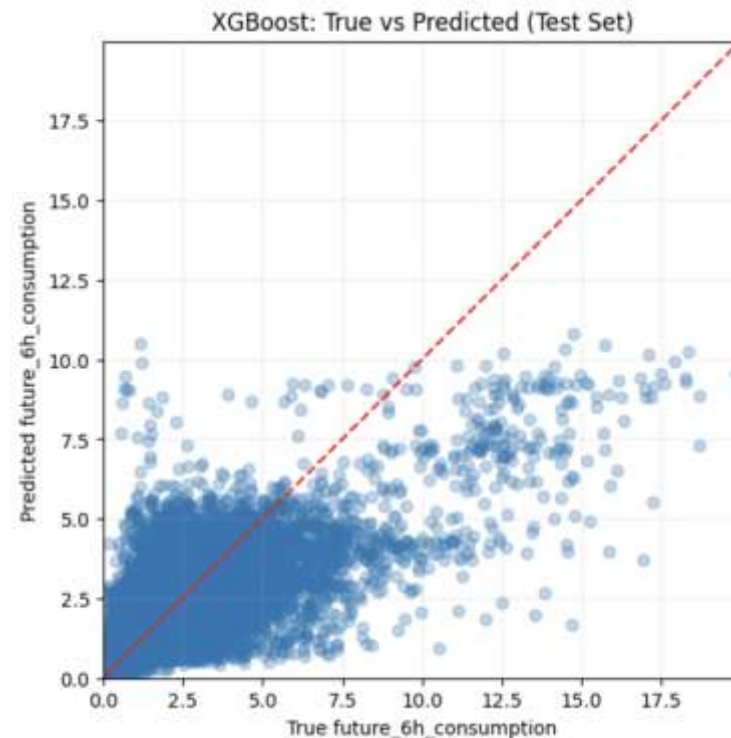
- Same features & preprocessing as Random Forest
- Key hyperparameters:
 - `n_estimators = 200`, `max_depth = 6`, `learning_rate = 0.1`
 - `subsample = 0.8`, `colsample_bytree = 0.8`
 - `objective = "reg:squarederror"`, `random_state = 42`

- **XGBoost Test Performance**

- MAE $\approx$ **0.3511**
- RMSE $\approx$ **0.6804**
- $R^2 \approx$ **0.6364**
- Trains **much faster** than the large Random Forest

- **Final Result:**

Model	MAE	RMSE	R^2
Random Forest (200×depth50)	0.3563	0.7145	0.5991
XGBoost (200, depth=6)	0.3511	0.6804	0.6364



Neural Networks

Presenter: Jinlei Chen

- MLP
 - As a baseline
 - Learns from static features
 - MAE=0.3795 | RMSE=0.7153 | R2=0.5982
- LSTM
 - Explore how time-based sequential models could help
 - Learns from historical consumption patterns
 - MAE=0.3525 | RMSE=0.6328 | R2=0.6877

Neural Networks

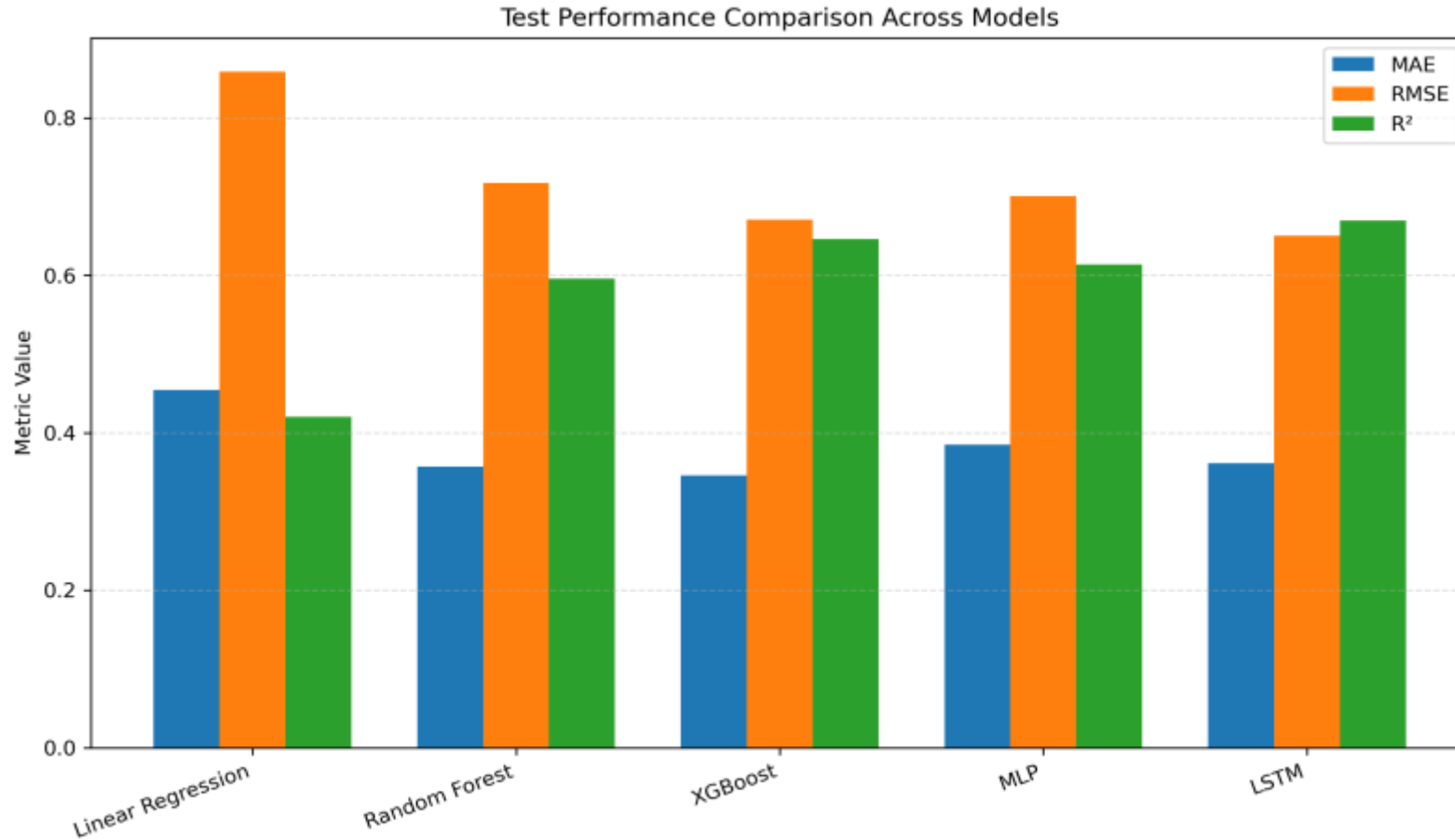
Presenter: Jinlei Chen

- The "Static Ceiling": MLP's performance matches the tree-based models, proving that static survey data alone has reached its predictive limit
- Temporal Superiority: LSTM outperforms MLP by capturing sequential dependencies and user habits that static features miss
- Data Strategy: Future improvements should focus on sequence length and granularity rather than adding more static survey questions.

Results, Discussion & Conclusion

Presenter: Xiaoyu Ma

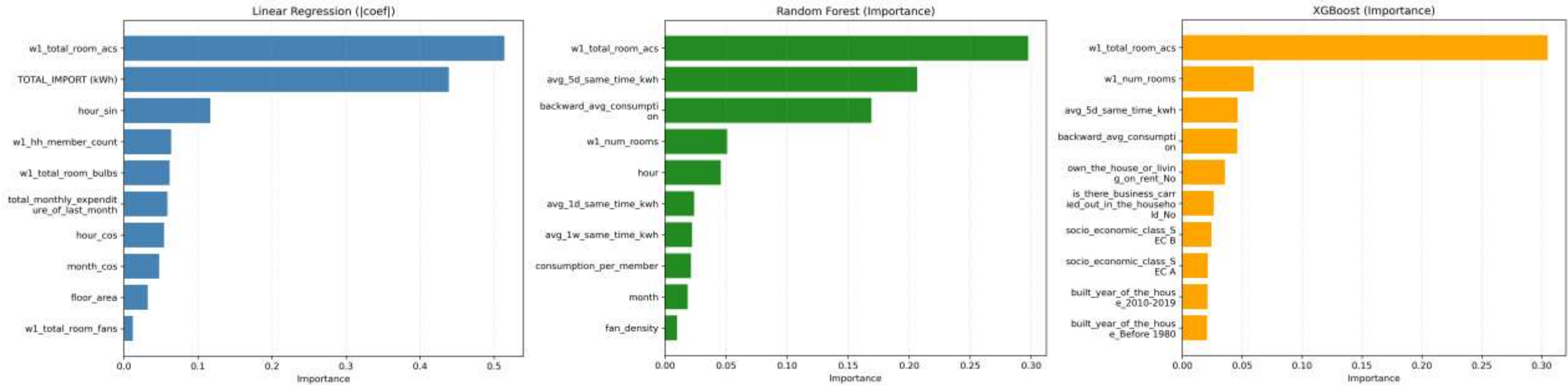
Comparison of MAE, RMSE, and R^2



Results, Discussion & Conclusion

Presenter: Xiaoyu Ma

Feature Importance

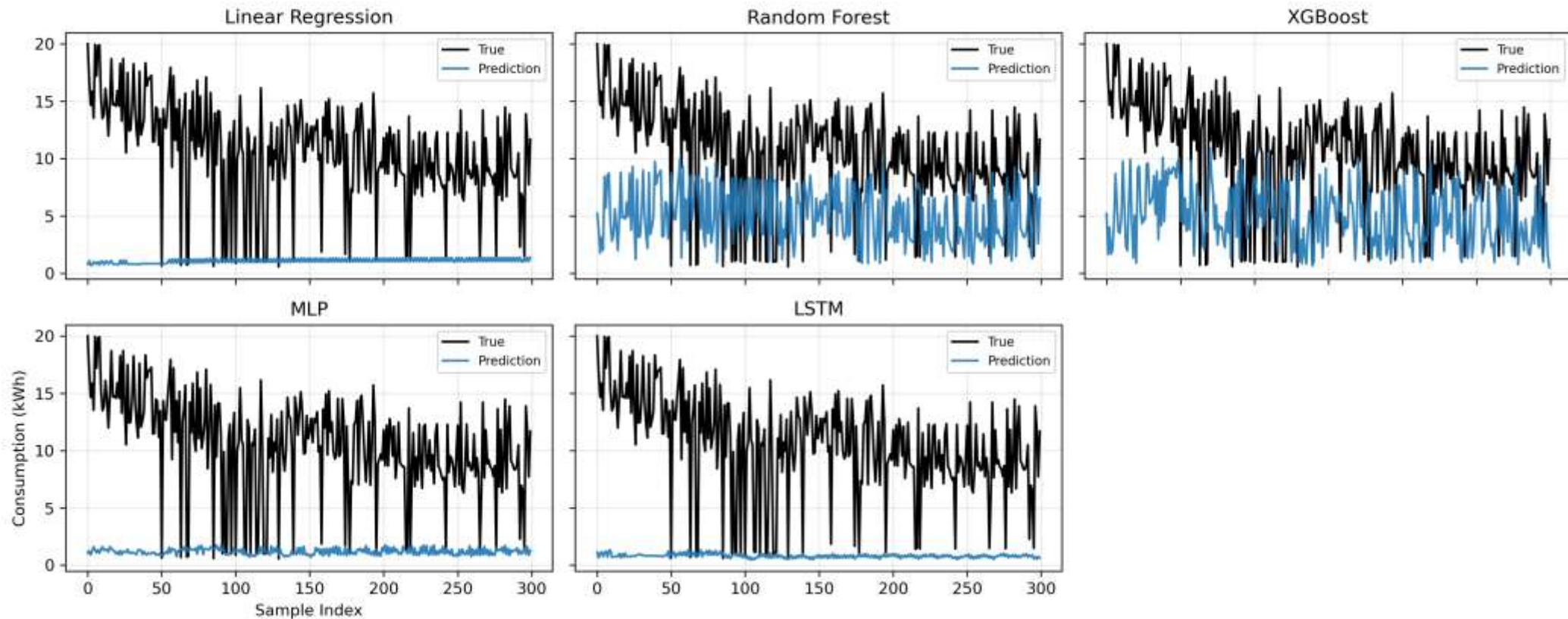


Results, Discussion & Conclusion

Presenter: Xiaoyu Ma

True vs. Predicted

True vs Predicted Future 6-hour Consumption (First 300 Test Samples)

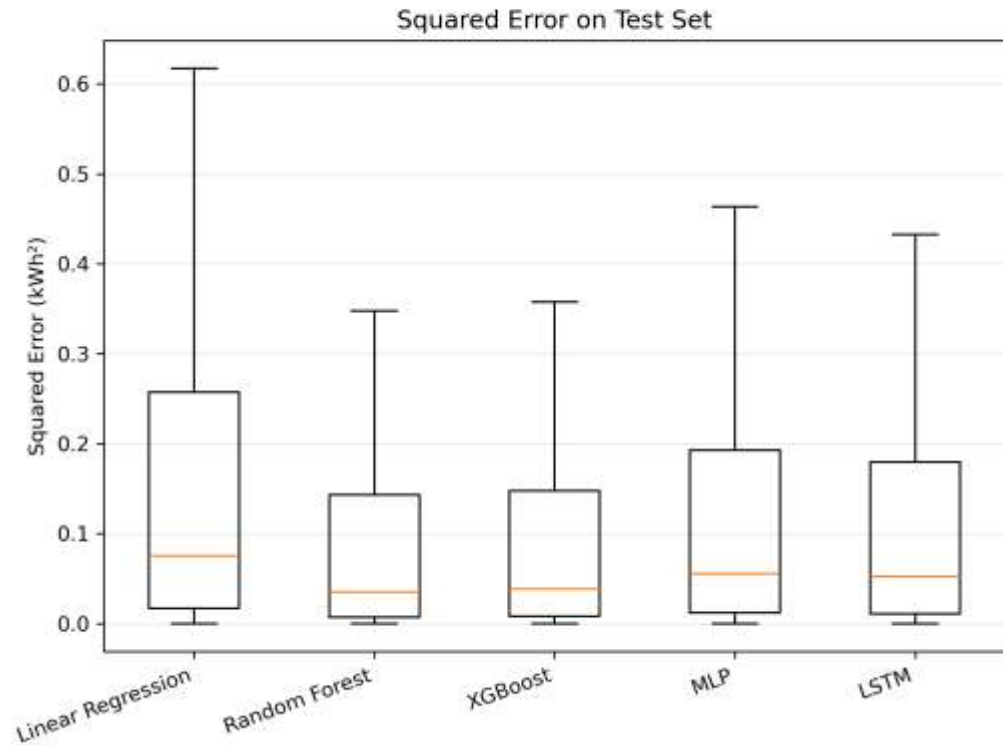
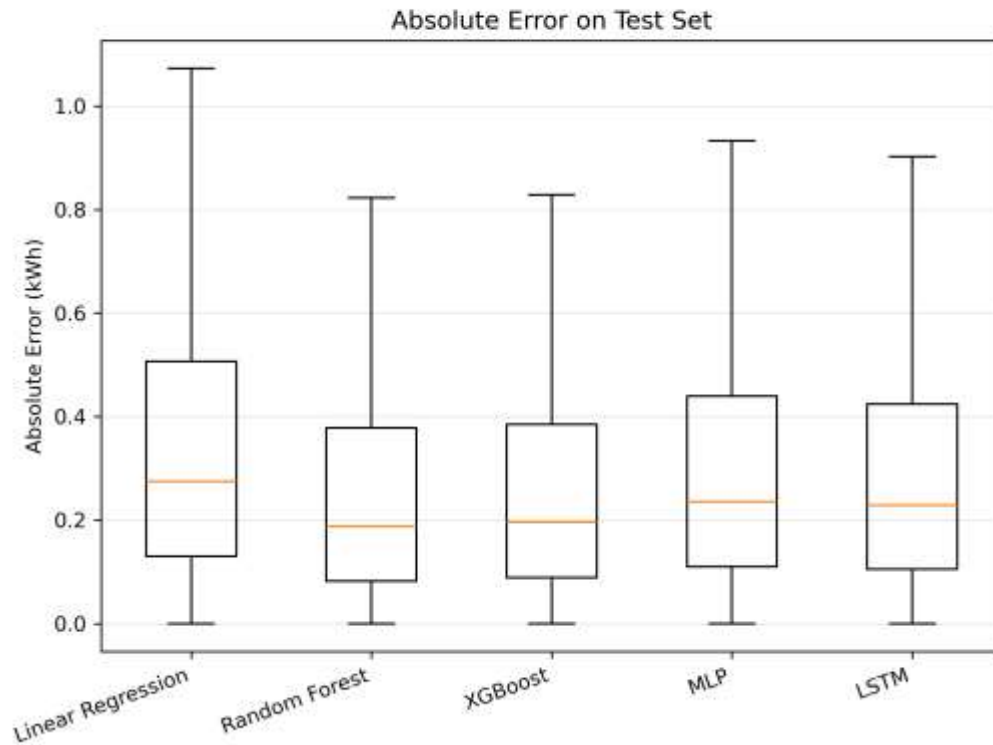


Results, Discussion & Conclusion

Presenter: Xiaoyu Ma

Test Error Analysis Across Models

Test Error Distribution Across Models



Thank You!

Group 2

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